

ATOMIC STRUCTURE

Particle	Charge	Relative charge	MASS	Relative Mass	Specific charge = $\frac{e}{m}$ ratio
Electron (J.J Thomson)	$-1.6 \times 10^{-19} \text{ C}$	-1	$9.1 \times 10^{-31} \text{ kg}$	$\frac{1}{1837}$	1837
Proton (Rutherford)	$+1.6 \times 10^{-19} \text{ C}$	+1	$1.6 \times 10^{-27} \text{ kg}$	1	1
Neutron (Chadwick)	zero	zero	$1.6 \times 10^{-27} \text{ kg}$	1	0
α -particle	$+2 \times 1.6 \times 10^{-19} \text{ C}$	+2	$4 \times 1.6 \times 10^{-27} \text{ kg}$	4	1/2

Learn :-

$$\begin{array}{ll}
 1 \text{ fm} = 10^{-15} \text{ m} & 1 \text{ nm} = 10^{-9} \text{ m} \\
 1 \text{ pm} = 10^{-12} \text{ m} & 1 \mu\text{m} = 10^{-6} \text{ m} \\
 1 \text{ \AA} = 10^{-10} \text{ m} & 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}
 \end{array}$$

Thomson's atomic model

(Plum pudding / Watermelon model)

- +ve charge $\rightarrow$ uniformly distributed
- -ve charge $\rightarrow$ embedded in cloud of the charge

$\rightarrow$ Limitation :- doesn't explain stability

Rutherford atomic model

- Most part of the atom is empty.
- All the +ve charge and mass of the atom are located in a very small space called nucleus at the centre of the atom.

* Important points :-

① $R_{\text{atom}} = 10^{-10} \text{ m}$, $R_{\text{nucleus}} = 10^{-15} \text{ m}$

$$R_{\text{atom}} = 10^5 R_{\text{nucleus}}$$

② Radius of nucleus of any atom,

$$R = R_0 [A]^{1/3}$$

$$\begin{aligned}
 R_0 &= 1.33 \times 10^{-13} \text{ cm} = 1.33 \times 10^{-15} \text{ m} \\
 &= 1.33 \text{ fermi}
 \end{aligned}$$

③ Nuclear density of all elements is same.

④ No. of α -particles deflected at angle $\theta \propto \frac{1}{[\sin \theta/2]^4}$

* Some Important terms

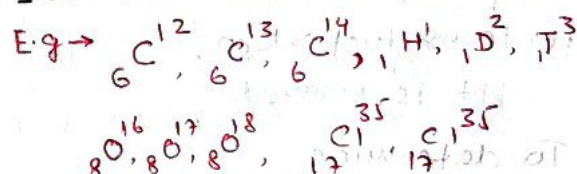
$A \rightarrow$ Mass number (P+N) or (Z+N)

${}_Z^X \rightarrow$ Symbol of element

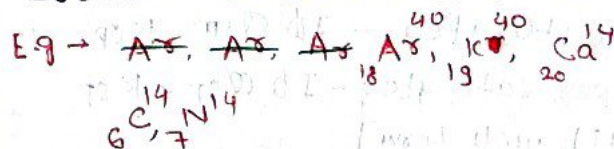
$\rightarrow$ Atomic no. = no. of protons
= no. of e^- (neutral)

$$A = Z + N$$

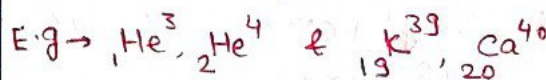
• Isotopes $\rightarrow Z \rightarrow$ same, $A \rightarrow$ different



• Isobars $\rightarrow Z \rightarrow$ different, $A \rightarrow$ same

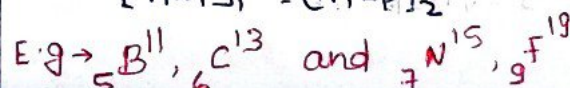


• Isotones $\rightarrow$ same no. of neutrons
($A - Z = N$)



• Isodiaphere $\rightarrow$ same difference of the no. of neutrons & protons.

$$[n - P]_1 = [n - P]_2$$

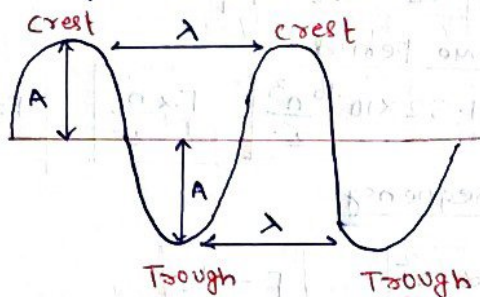


- **Isostere** → Same no. of atoms & electrons.
E.g → $\text{CO}_2, \text{N}_2\text{O} \& \text{CaO, KF}$
- **Isoelectronic** → Same no. of electrons.
E.g → $(\text{Cl}^-, \text{Ar}), (\text{NO}^+, \text{CO}), (\text{H}_2\text{O}, \text{NH}_3)$

Electromagnetic Radiation / Wave

- **Wavelength (λ)** $\lambda = \frac{c}{\nu}$

↳ distance b/w 2 crest or trough or length of one wave.



- **Amplitude (A)** :- It expresses the maximum displacement by wave from equilibrium state.
- **Frequency (ν)** :- (Unit = Hz)
- **Speed (c)** :- $c = \nu \lambda$ [$c = 3 \times 10^8 \text{ m/s}$]
- **Wave number ($\bar{\nu}$)** :- $\bar{\nu} = \frac{1}{\lambda}$
- **Time period (T)** :- $T = \frac{1}{\nu}$

Electromagnetic spectrum

↓ Cosmic rays	($\lambda \uparrow, \nu \downarrow, E \downarrow$)	TRICK
↓ Gamma rays		classic
↓ X-rays		radiation
↓ U.V radiation		UV
↓ visible → VIBGYOR	$\lambda \uparrow, \nu \downarrow$	V
↓ Infrared		In
↓ Microwave		Mγ
↓ Radiowaves		Range

Planck's Quantum theory

- Energy is emitted discontinuously in packets of photons

• $E \propto \nu$ $E = h\nu$

↳ $E = \frac{hc}{\lambda}$ for 1 quantum

$E = \frac{nhc}{\lambda}$ for n quantum

$E = n \frac{12400 \text{ eV}}{\lambda(\text{\AA})}$

$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$
 $h = 6.6 \times 10^{-34} \text{ Js}$
 $n = \text{no. of photons}$
 $h = 6.6 \times 10^{-27} \text{ erg.s}$

Photoelectric effect

- When electrons are ejected from the surface of a metal when light is incident on it.
- Example of particle nature
- Surface phenomena
- Single photon release single e^-
- No time gap.

$E = \phi + K.E_{\text{max}}$ $\omega_0 \rightarrow \text{work function}$
 $\downarrow$
 $\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + K.E_{\text{max}}$ $\nu = \text{incident freq.}$
 $\nu_0 = \text{Threshold freq.}$
 $K.E = h(\nu - \nu_0)$

$\nu > \nu_0 \rightarrow e^-$ emitted with $K.E$ ($\lambda < \lambda_0$)

$\nu = \nu_0 \rightarrow e^-$ emitted without $K.E$ ($\lambda = \lambda_0$)

$\nu < \nu_0 \rightarrow e^-$ not emitted ($\lambda > \lambda_0$)

* Black body Radiation

$T \uparrow \uparrow, \lambda \downarrow \downarrow$

Telegram:- iitjeeenotess

* Stopping potential (V_0)

$$K.E_{\max} = eV_0$$

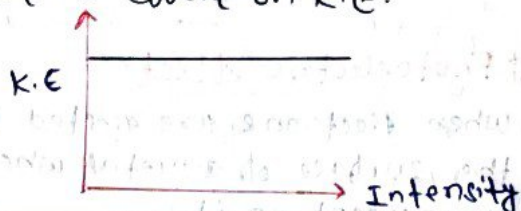
$$V_0 = \frac{h\nu - h\nu_0}{e}$$

• Einstein photoelectric equation

$$h\nu = h\nu_0 + \frac{1}{2} m_e v^2$$

Intensity of light

- Total no. of photon incident / Area / second
- Intensity $\uparrow$, no. of photons $\uparrow$
no. of photoelectrons $\uparrow$, photoelectric current $\uparrow$
 $\rightarrow$ but no effect on K.E.



Bohr's Atomic Model

- only applicable for single e^- species like (H, He^+, Li^{+2}) etc.
- e^- revolve only in those orbits where its angular momentum is an integer multiple of $\frac{h}{2\pi}$.

$$mvr = \frac{nh}{2\pi} \quad n=1, 2, 3, 4, \dots$$

$\rightarrow$ These are called stationary orbits. In orbit, e^- does not release any energy.

$$\frac{mv^2}{R} = \frac{KZe^2}{R^2}$$

- The emission or absorption of energy in the form of photon can only occur when electron jumps from one stationary state to another.

$$\Delta E = E_{\text{higher}} - E_{\text{lower}}$$

$$\Delta E = E_{n_2} - E_{n_1}$$

* Radius of n th Bohr's orbit

$$r_n = 0.53 \times \frac{n^2}{Z} \text{ \AA}$$

$$r_n \propto \frac{n^2}{Z}$$

$$\frac{r_1}{r_2} = \frac{n_1^2}{n_2^2} \frac{Z_2}{Z_1}$$

* Velocity of n th orbit

$$V_n = 2.18 \times 10^6 \times \frac{Z}{n}$$

$$V \propto \frac{Z}{n}$$

$$\frac{V_1}{V_2} = \frac{Z_1}{Z_2} \frac{n_2}{n_1}$$

* Time period

$$T = 1.52 \times 10^{-16} \frac{n^3}{Z^2}$$

$$T \propto \frac{n^3}{Z^2}$$

$$T = \frac{2\pi r}{V}$$

* Frequency

$$f \propto \frac{Z^2}{n^3}$$

$$f = \frac{1}{T}$$

* Energy of an electron n th Bohr's orbit

$$E = -13.6 \frac{Z^2}{n^2} \text{ eV/atom}$$

$$E = -313.6 \frac{Z^2}{n^2} \text{ kcal/mole}$$

$$1 \text{ eV} = 3.83 \times 10^{-23} \text{ kcal}$$

$$1 \text{ eV} = 1.6 \times 10^{-12} \text{ erg}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$1 \text{ cal} = 4.18 \text{ J}$$

- Energy difference b/w n_1 & n_2 energy level

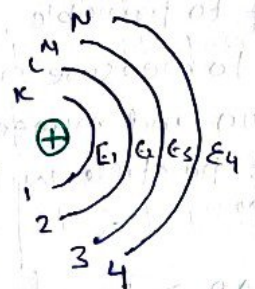
$$\Delta E = E_{n_2} - E_{n_1}$$

$$= 13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV/atom}$$

Telegram:- iitjeeneetnotess

• Energy level of H-atom can be represented as follow:-

$n=1$ $E_1 = -13.6 \text{ eV}$, $E_2 - E_1 = 10.2 \text{ eV}$
 $n=2$ $E_2 = -\frac{13.6}{4} = -3.4 \text{ eV}$, $E_3 - E_2 = 1.89 \text{ eV}$
 $n=3$ $E_3 = -\frac{13.6}{9} = -1.51$, $E_4 - E_3 = 0.66 \text{ eV}$
 $n=4$ $E_4 = -\frac{13.6}{16} = -0.85 \text{ eV}$, $E_5 - E_4 = 0.31 \text{ eV}$



• T.E = $-13.6 \times \frac{Z^2}{n^2} \text{ eV}$

• K.E = $+13.6 \times \frac{Z^2}{n^2} \text{ eV}$

• P.E = $-27.2 \times \frac{Z^2}{n^2} \text{ eV}$

PE = -2KE

KE = -TE

PE = 2TE

TE = -TE = $|-13.6 \times \frac{Z^2}{n^2}|$

K.E = $-\frac{PE}{2} = -TE$

KE : PE : TE = 1 : -2 : -1

Hydrogen spectrum

Telegram:- iitjeeenotess

Series	n_1	n_2	Spectral Region
Lyman series	1	2, 3, 4, 5, ...	U.V (ultraviolet) rays
Balmer series	2	3, 4, 5, 6, ...	Visible rays
Paschen series	3	4, 5, 6, 7, ...	Near Infrared (I.R)
Brackett series	4	5, 6, 7, 8, ...	I.R
Pfund series	5	6, 7, 8, 9, ...	far I.R

Calculation of no. of spectral lines

• Total no. of spectral lines = $\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$

• If $n_1 = 1$ (ground level)

Total no of spectral lines = $\frac{n(n-1)}{2}$

Rydberg formula

↳ wave length of emitted line

$\bar{\nu} = \frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

↳ wave numbers

R = Rydberg constant

$R = 109678$

De Broglie's hypothesis

↳ Matter shows both particle & wave like properties.

$\lambda = \frac{h}{p} = \frac{h}{mv}$

$\lambda = \frac{h}{\sqrt{2mKE}} = \frac{h}{\sqrt{2mqV}}$

$\lambda = \frac{12.24 \text{ \AA}}{\sqrt{V}} = \sqrt{\frac{150}{V}} \text{ \AA}$ → For electron
→ volt

$2\pi r n = n\lambda$ ≠ no. of wavelength require to complete

One cycle in nth orbit = n.

Telegram:- iitjeeenotess

* Heisenberg Uncertainty principle

↳ According to principle, It is impossible to measure simultaneously the position and momentum of a microscopic particle with absolute accuracy.

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \cdot m \Delta v \geq \frac{h}{4\pi}$$

$$\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi} \quad \frac{h}{4\pi} = 0.527 \times 10^{-34} \text{ J sec}$$

Δx = uncertainty in position

Δp = " " " momentum

Δv = " " " velocity

m = mass of particle

h = Planck's constant

Quantum number

① Principle Q.N (n)

↳ It represents name, size & Energy of the shell to which e^- belongs.

• Energy $\uparrow$, $n \uparrow$

• $n = 1, 2, 3, 4$

↓ K L M N
shell

• Total no. of electrons in a particular shell (orbit) = $2n^2$.

• Total no. of ~~electrons~~ orbitals in an orbit = n^2 .

• Total no. of subshell in an orbit = n

• Total no. of value of 'l' = 'n'

$n = 3$, $l = 0, 1, 2$

$n = 4$, $l = 0, 1, 2, 3$

② Azimuthal Q.N (l):-

↳ It represents the shape of the subshell & orbital.

• orbital angular momentum

$$= \sqrt{l(l+1)} \frac{h}{2\pi} \text{ or } \sqrt{l(l+1)} \hbar$$

$$(\because \hbar = \frac{h}{2\pi})$$

• Value of 'l' is b/w 0 to $(n-1)$

→ value of 'l' Subshell

$l = 0$ — s (sharp)

$l = 1$ — p (principal)

$l = 2$ — d (diffused)

$l = 3$ — f (fundamental)

• Total no. of electron in a sub-orbit = $2(2l+1)$

• Total no. of orbital in a sub-orbit = $(2l+1)$

③ Magnetic Q.N (m)

• Orientation of orbital

• Value of $m \rightarrow -l$ to $+l$ including zero.

↳ value of 'l' value of 'm'

0 — 0
1 — -1, 0, +1
2 — -2, -1, 0, +1, +2

④ Spin Quantum number (s)

↳ It represents the direction of electron spin around its own axis.

• Value of $s = +\frac{1}{2}$ or $-\frac{1}{2}$

↙ clockwise ↘ Anticlockwise

• Total spin = no. of unpaired electron $\times \pm \frac{1}{2}$

• Spin angular momentum

$$= \sqrt{s(s+1)} \frac{h}{2\pi} \text{ or } \sqrt{s(s+1)} \hbar$$

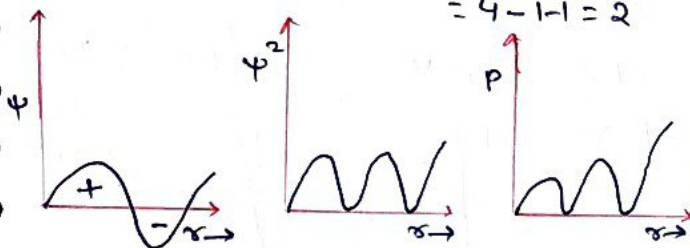
Node

- no. of Radial node = $n-1-l$
- no. of Angular node = l
- total node = $n-1$

Step 1 $\rightarrow$ no. of radial node = $n-1-l$

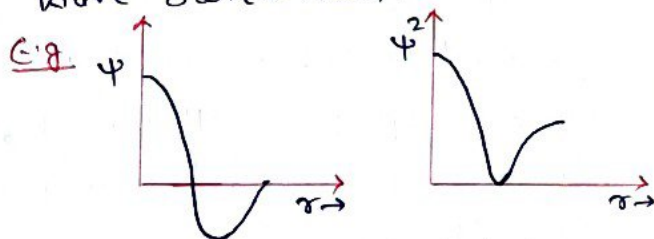
Step 2 $\rightarrow$ no. of radial node $\rightarrow$ no. of times $\psi = 0$

E.g. $\rightarrow 4p \rightarrow$ Radial node = $n-1-l$
 $= 4-1-1 = 2$



$\rightarrow$ Graph touch the line 2 times.

- only 's' orbital ka graph (ψ, ψ^2) upar se ata hai, baaki sab ka wave banta hai.



Rules for filling orbitals

① Aufbau principle

- e^- filled in increasing order of subshell.
- $1s^2 < 2s^2 < 2p^6 < 3s^2 < 3p^6 < 4s^2 < 3d^{10} < 4p^6 < 5s^2 < 3f^{14} < 4d^{10} < 5p^6$

$d^4 \rightarrow d^5$
 $d^9 \rightarrow d^{10}$ } more stable

violation $\rightarrow$

1

1	1	1
---	---	---

② (n+1) Rule

- 1. $\rightarrow$ Energy of subshell $\propto (n+1)$ value of subshell.

If rule can't calculate energy of subshell then,

2. $\rightarrow$ Energy of subshell $\propto$ value of 'n' of subshell.

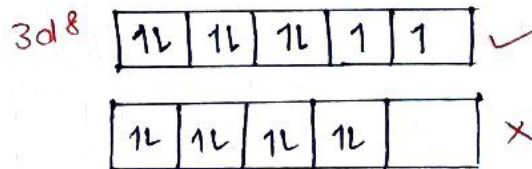
Shell $\rightarrow n \leq l \rightarrow$ not possible
 $l_{\max} = n-1, l_{\min} = 0$

③ Pauli's exclusion principle

- $\rightarrow$ No two electrons in atom can have same set of 4 quantum no.
- $\rightarrow$ orbital can accommodate max. 2 electron with opposite spin.

④ Hund's rule

- $\rightarrow$ pairing of electrons in orbitals belonging to same subshell does not take place until each orbital belonging to that subshell has got one electron.



Magnetic moment

$$\mu = \sqrt{n(n+2)}$$

$\therefore n =$ unpaired electron in sub-shell.

Telegram:- iitjeeneetnotess